HOUSE BILL 2316

By Barrett

AN ACT to amend Tennessee Code Annotated, Title 4; Title 7 and Title 65, relative to threats to electrical assets.

BE IT ENACTED BY THE GENERAL ASSEMBLY OF THE STATE OF TENNESSEE:

SECTION 1. As used in this act:

(1) "Covered entity" means an electric utility or electric project developer, owner, or operator within this state, regardless of public or private ownership or jurisdiction under the North American Electric Reliability Corporation (NERC);

(2) "Covered equipment" means any power transformer with a primary voltage of one hundred kilovolts (100 kV) or greater and capacity of twenty-five megavolt ampere (25 MVA) or greater, and any generator step-up transformer with a secondary voltage of one hundred kilovolts (100 kV) or greater and capacity of twenty-five megavolt ampere (25 MVA) or greater;

(3) "GIC" means a geomagnetically induced current, also known as ground induced current, resulting from naturally occurring geomagnetic disturbances or the late-time component of a high-altitude nuclear electromagnetic pulse; and

(4) "Critical energy infrastructure information protocols," "critical electric infrastructure information protocols," or "CEII" means specific engineering, vulnerability, or detailed design protocols and procedures related to proposed or existing critical infrastructure, whether physical or virtual, that relate to the production, generation, transmission, transportation, or distribution of energy, the unauthorized disclosure of which could pose a risk to the security, reliability, or integrity of the infrastructure; such protocols are designated as confidential and exempt from public disclosure, as their

release could be useful to a person planning an attack or otherwise causing harm to the infrastructure.

SECTION 2.

(a) Each covered entity shall, no later than January 1, 2027, conduct a technical assessment of all covered equipment to determine vulnerability to GICs.

(b) The assessment shall:

(1) Utilize the waveform in Figure A.5 of IEC 61000-2-9, Edition 2.0 (2025-05), modeling a peak magnetic field strength of twenty thousand nanotesla (20,000 nT) and corresponding electric field of eighty-five volts per kilometer (85 V/km);

(2) Assume transformers are fully loaded during GIC exposure;

(3) Account for transformer age and condition using ANSI/IEEE Standard C57.110 and IEEE Standard C57.91; and

(4) Identify susceptibility to half-cycle saturation, GIC-induced harmonics, reactive power consumption, hot spot generation, and insulation degradation.

SECTION 3.

(a) No later than July 1, 2027, each covered entity shall submit a report on the assessment conducted pursuant to Section 2 to the general assembly, the governor, the Tennessee board of utility regulation, and the Tennessee public utilities commission.

(b) The report must include, for each susceptible transformer and substation:

(1) The transformer:

(A) Brand;

(B) Place of origin, including nation where manufactured;

(C) Design specifications, including windings and core configuration; winding impedances; winding DC resistances, specifying whether assumed or measured; and phase type, either single-phase or 3-phase;

(D) Capacity in megavolt-amperes (MVA);

(E) Voltage level in kilovolts (kV);

(F) Age;

(G) Site location, redacted for CEII;

(H) Purpose, such as generator step-up, autotransformer, step-down, converter, redacted for CEII;

(I) Replacement lead time; and

(J) Replacement cost;

(2) Spreadsheet-formatted data and narrative analysis;

(3) Recommended solutions to protect the grid against GIC by preventing or reducing the half-cycle saturation of transformers;

(4) Total cost to implement GIC protection;

(5) Priority list of transformers by damage risk and critical infrastructure impact; and

(6) Funding recommendations, including potential grant sources and rate recovery mechanisms.

SECTION 4. Covered entities shall not rely solely on operational procedures such as load shedding or reactive power supply to mitigate GIC risk. Such procedures are not considered sufficient protection under this chapter.

SECTION 5. CEII data, such as location and purpose, submitted under this act must be handled in accordance with CEII protocols as defined in Section 1. Location and purpose data must be redacted from public reports.

SECTION 6. This act takes effect upon becoming a law, the public welfare requiring it.